

BEVERAGE SUPPLY CHAIN & INVENTORY OPTIMIZATION

Systematic Data Analysis Report

Supplier relationships • Inventory movement • Product performance • Operational risk

1. Executive Summary

This project analyzes beverage supply-chain activity across retail sales, retail transfers, and warehouse sales to identify opportunities for improving supplier management, inventory movement, and product-level performance.

The dataset contains 330,080 records and 9 variables. Performed data cleaning, exploratory analysis, activity classification, supplier analysis, category movement analysis, dead-stock identification, trend analysis, supplier concentration analysis, supplier specialization, distribution analysis, skewness analysis, and outlier analysis.

The analysis shows that warehouse activity is the dominant component. Multiple-activity records account for approximately 43.49% of records, while warehouse-sales-only records account for approximately 38.53%. Beer is the largest-moving category within multiple-activity records, at approximately 6.55 million units.

Supplier concentration is a key operational-risk finding. The top 20 suppliers account for approximately 67.49% of retail sales, 79.16% of warehouse sales, and 67.82% of retail transfers.

2. Business Problem

A large beverage distribution company manages thousands of products across categories such as Beer, Wine, and Liquor. Products are purchased from hundreds of suppliers and distributed through multiple channels.

Retail Sales represent products sold through retail operations. Retail Transfers represent movement between retail locations or channels. Warehouse Sales represent products moved or sold through the warehouse distribution channel.

The central business question is how the company can optimize supplier relationships, improve inventory movement, strengthen product-level sales performance, and reduce supplier dependency and operational risk.

3. Project Objectives

- Identify major suppliers and understand their contribution to retail and warehouse activity.
- Measure supplier concentration and identify potential dependency risks.
- Understand supplier specialization by product category.
- Compare retail sales, retail transfers, and warehouse sales.
- Identify high-volume product categories and inventory-movement patterns.
- Identify inactive or potentially dead-stock records.
- Examine trends over the available years.
- Use distribution, skewness, and outlier analysis to understand product-level sales behavior.

4. Dataset Overview

Dataset size: 330,080 rows × 9 columns.

Variables: Year, Month, Supplier, Item Code, Item Description, Item Type, Retail Sales, Retail Transfers, and Warehouse Sales.

5. Data Cleaning & Preparation

Column names were standardized to lowercase. Retail Sales, Retail Transfers, and Warehouse Sales were cleaned by removing commas, converting parenthesized values to negatives, and converting the fields to numeric data types.

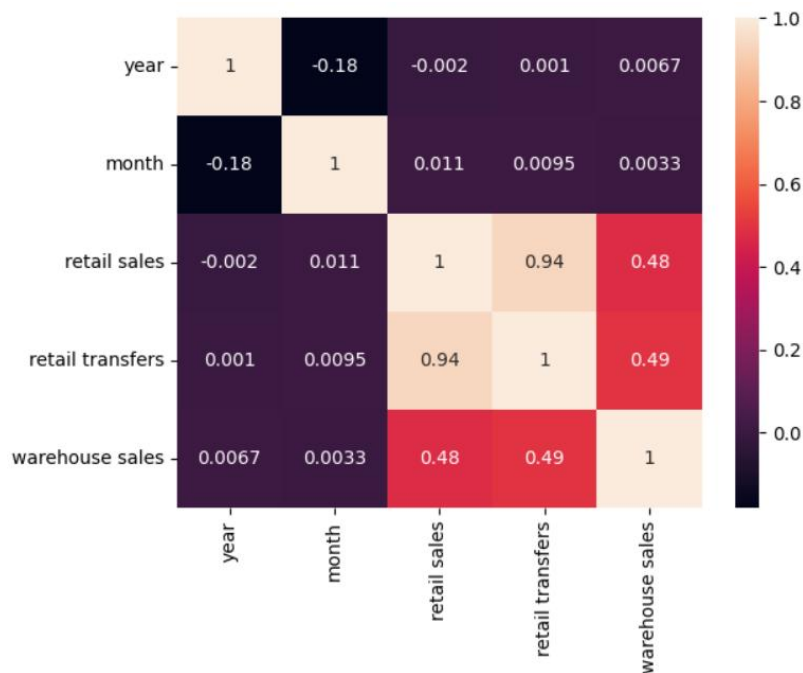
Missing supplier, item-type, and retail-sales values were identified and were handled. Negative values were retained because they can represent returns, reversals, corrections, or inventory adjustments.

Negative retail transfers occurred in approximately 0.31% of records and should be investigated rather than automatically removed.

6. Exploratory Data Analysis

A strong positive relationship was observed between retail sales and retail transfers. Products with higher retail sales activity tend to experience higher transfer activity.

Many records participate in only one movement channel, so retail sales should not be analyzed in isolation.



7. Inventory Activity Classification

Multiple Activities: 143,548 records (43.489%).

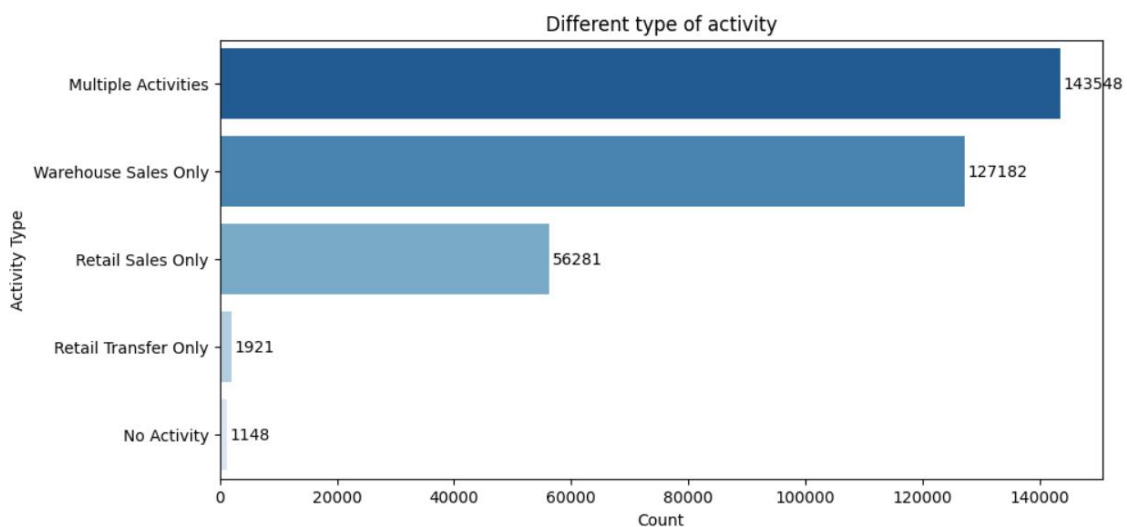
Warehouse Sales Only: 127,182 records (38.531%).

Retail Sales Only: 56,281 records (17.051%).

Retail Transfer Only: 1,921 records (0.582%).

No Activity: 1,148 records (0.348%).

More than 43% of records involve multiple types of activity, demonstrating the importance of analyzing the full inventory-movement cycle.



8. Supplier Analysis

E & J GALLO WINERY was identified as the top retail-sales supplier. CROWN IMPORTS was identified as the top warehouse-sales supplier.

Supplier importance differs by distribution channel, so supplier evaluation should consider retail contribution, warehouse contribution, transfer activity, product breadth, and consistency.

9. Inventory Movement by Product Category

Beer: 6,551,141.88 total movement.

Wine: 2,473,875.26.

Liquor: 1,834,433.65.

Dunnage: 129,361.00.

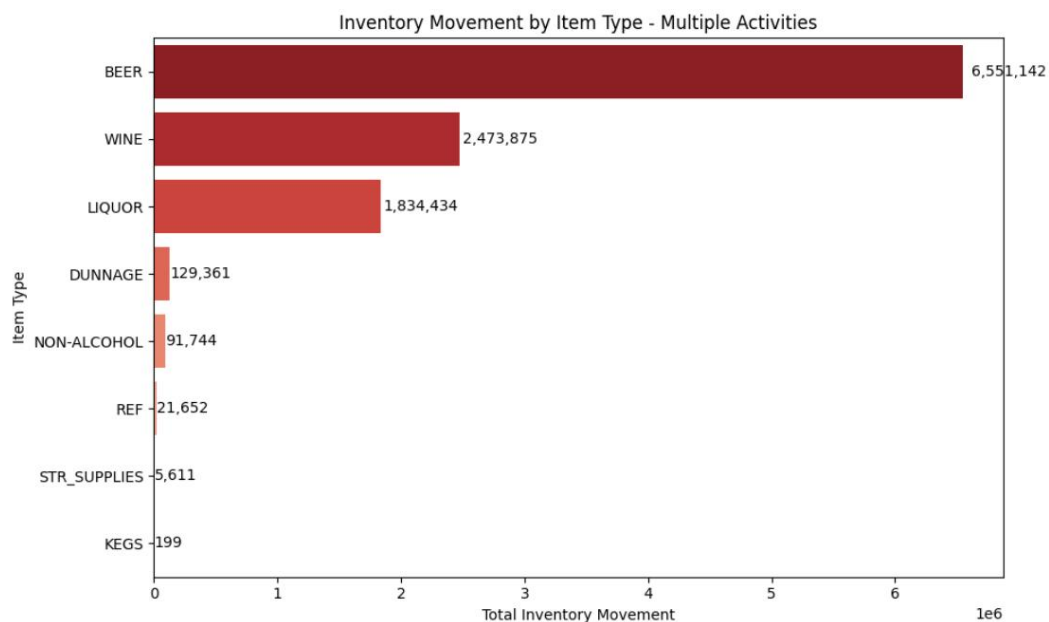
Non-Alcohol: 91,743.55.

REF: 21,651.91.

STR_SUPPLIES: 5,611.09.

Kegs: 199.00.

Beer represents approximately 58.98% of total movement in the multiple-activity analysis.



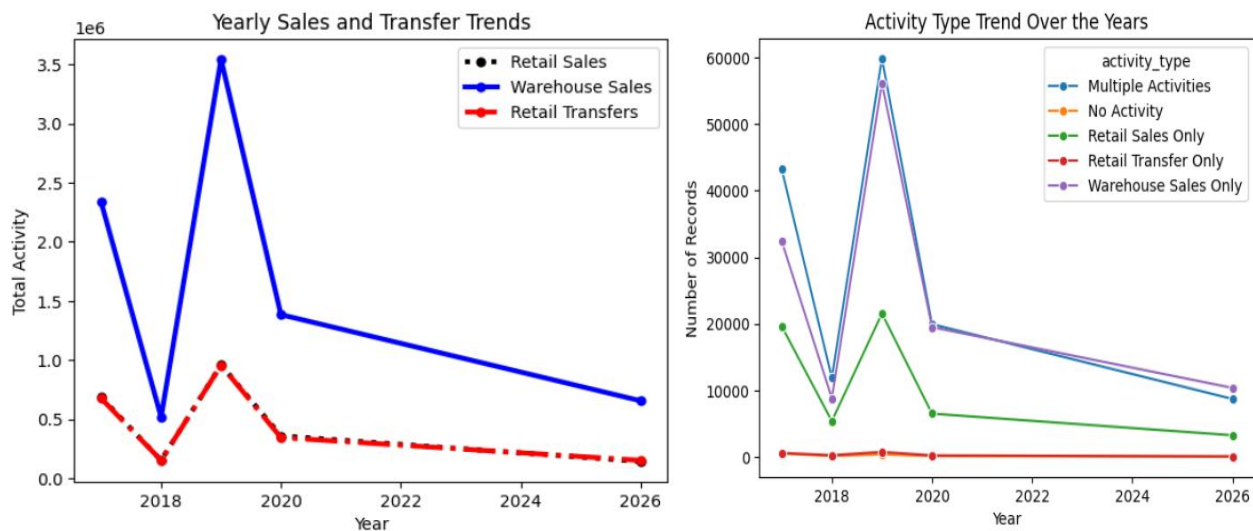
10. Dead Stock / Inactive Inventory

Identified 1,148 records with zero retail sales, zero retail transfers, and zero warehouse sales. Inactive records by major item type include Wine (767), Beer (164), Liquor (106), and Kegs (102).

These records may represent slow-moving products, obsolete products, discontinued items, temporary inactivity, or data-quality issues.

11. Yearly Trend Analysis

Warehouse sales dominate yearly activity. Among the available years, warehouse sales show substantial fluctuations and peak in 2019.



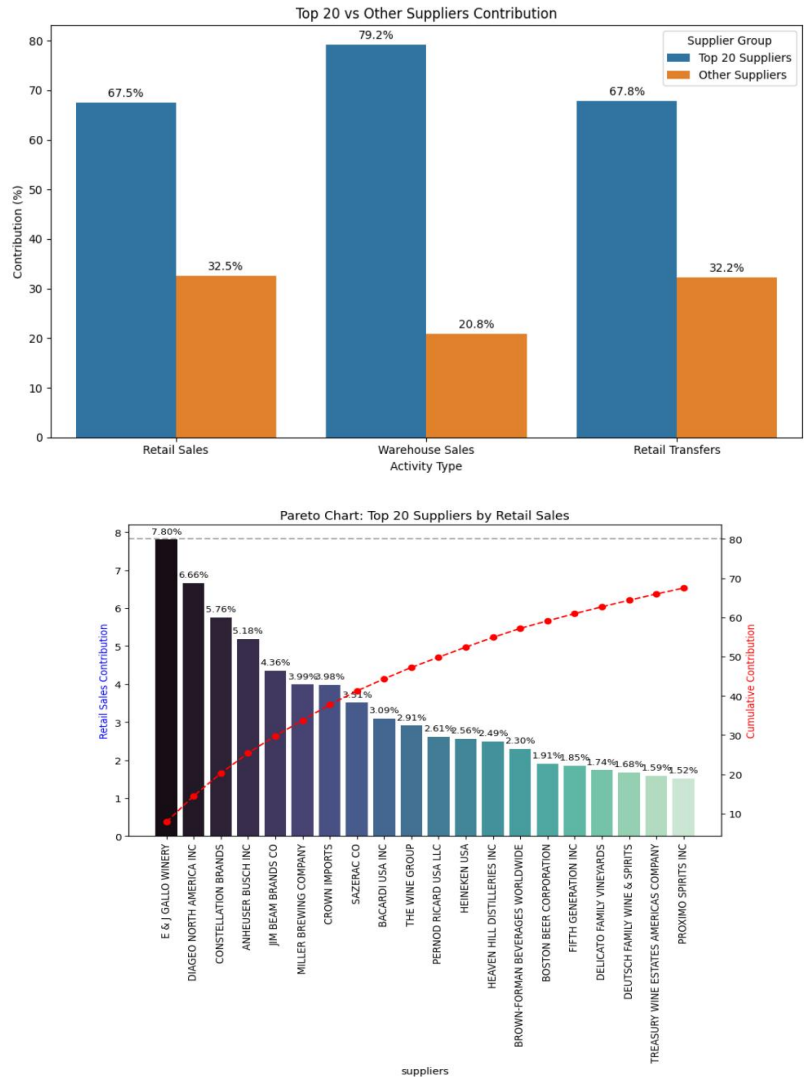
The available data does not provide continuous annual coverage from 2021 through 2025, so long-term trend claims should be interpreted cautiously.

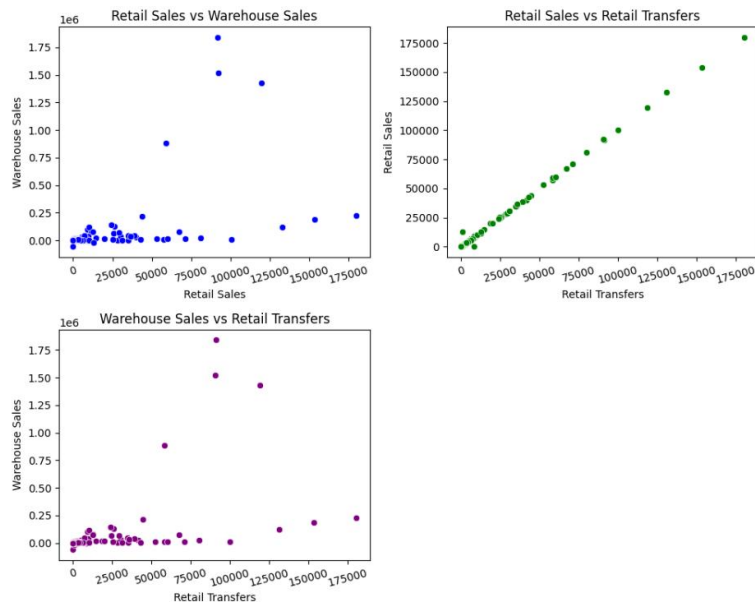
12. Supplier Concentration & Risk

Top 20 suppliers account for approximately 67.49% of retail sales, 79.16% of warehouse sales, and 67.82% of retail transfers.

The strongest concentration occurs in warehouse sales. Major suppliers also show strong category specialization across Beer, Wine, and Liquor.

Supplier dependency should therefore be assessed at both supplier level and supplier-plus-category level.



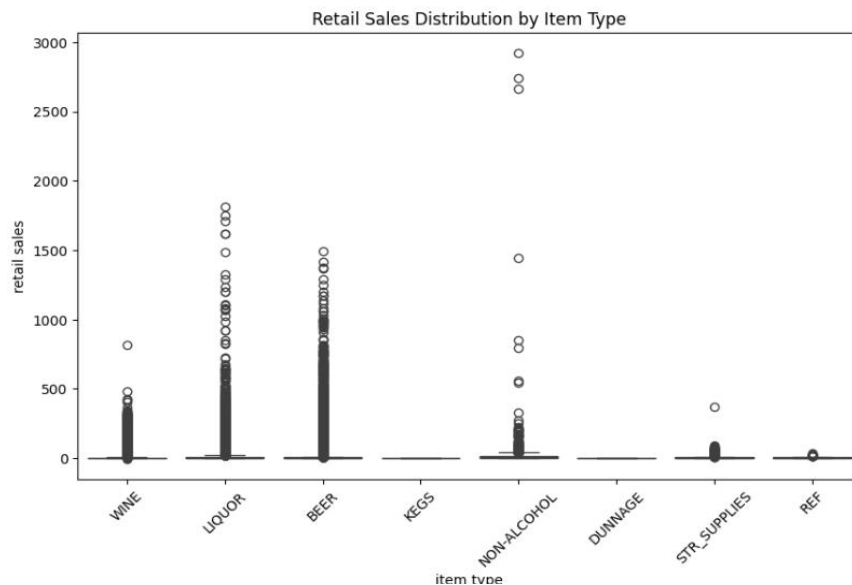
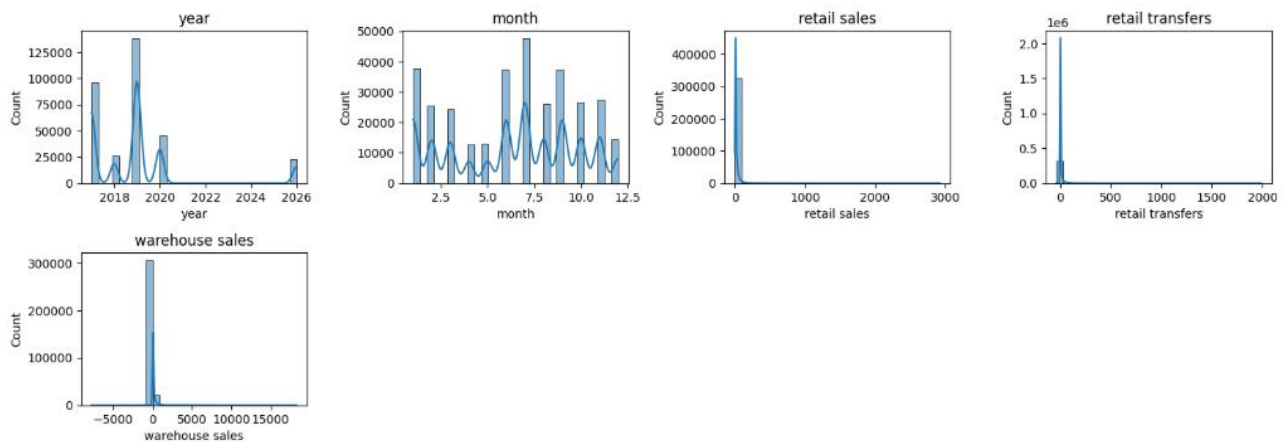


13. Distribution, Skewness & Outlier Analysis

Retail sales are highly right-skewed, with many zero or low observations and a relatively small number of very high-sales observations.

Transfer values show a similar pattern, with most observations close to zero and a smaller number of large movements.

High-sales observations should not automatically be removed because they may represent successful products, bulk orders, promotions, seasonal demand, or one-time events.



14. Key Business Findings

- Warehouse activity dominates overall inventory movement.
- Multiple-channel activity is common.
- Beer is the largest-moving category.
- Supplier concentration is significant, especially in warehouse sales.
- Supplier specialization creates category-level dependency risk.
- 1,148 inactive records warrant investigation.
- Sales are highly skewed, making segmentation and outlier analysis important.
- Negative movements are rare but should be reviewed.

15. Business Recommendations

- Develop a supplier risk matrix using sales contribution, warehouse contribution, product breadth, category concentration, and alternative-supplier availability.
- Diversify critical suppliers for strategically important products.
- Negotiate better purchase prices, freight costs, volume discounts, payment terms, and service-level agreements with strategic suppliers.
- Prioritize Beer for demand forecasting, replenishment, warehouse capacity planning, and supplier monitoring.
- Investigate inactive products and classify them as temporary inactivity, slow-moving, obsolete, discontinued, or data-quality issues.
- Investigate negative transactions for returns, reversals, corrections, and adjustments.
- Implement product segmentation using sales and movement levels.

16. Limitations & Future Analysis

Currently it is primarily a movement and exploratory analysis. Profitability requires purchase cost, selling price, gross profit, and margin.

Bulk-purchasing analysis requires purchase quantity, purchase price, order quantity, and unit cost.

Inventory-turnover analysis requires opening/closing or average inventory.

Additional supplier-risk analysis would benefit from lead times, service levels, alternative suppliers, and contract information.

17. Conclusion

The beverage distribution business has a complex inventory network in which warehouse activity plays a particularly important role. Beer is the dominant movement category, while a relatively small group of suppliers contributes a very large share of activity.

The strongest opportunities are supplier-risk management, supplier diversification, strategic negotiation, warehouse optimization, Beer-category planning, dead-stock reduction, and product segmentation.

With additional purchase-cost, profitability, and inventory-level data, this project can be extended into a complete supplier-profitability, inventory-turnover, and supply-chain optimization solution.